Solution Requirements

Date	20 October 2026
Team ID	Individual project (no team)
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Version 1 is a public demo and needs no login. Supabase email authentication is planned for Phase 2 so trips belong to a user.	Medium
2	Authorization levels	Single role in version 1: any visitor can create, save, view and delete trips. Per-user access with row level security is planned.	Medium
3	External interfaces	Groq API (LLM generation, JSON mode) and Supabase (PostgreSQL via supabase-py). Frontend calls the FastAPI backend over HTTPS.	High
4	Transactions processing	Generate itinerary (POST /itinerary); save trip (POST /trips inserts a row); list trips (GET /trips); delete trip (DELETE /trips/{id}).	High
5	Reporting	Day-wise itinerary with timings, travel times, estimated cost per stop and total; copy itinerary as text; saved trips list.	Medium
6	Business rules, etc.	Destination is mandatory; days between 1 and 14; budget is low, medium or premium; vibe is relaxed, adventure, romantic or cultural; nearby places are grouped per day and ordered by time.	High
7	Compliance to laws or regulations	No personal data is collected. API keys are kept only in server environment variables and never exposed in the frontend or the repository.	Medium
8	Other	Destination suggestions (POST /recommend) based on budget, interests and vibe.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Non-AI endpoints respond quickly; itinerary generation depends on the LLM.	Non-AI endpoints < 2 s; POST /itinerary < 20 s
2	Scalability	Stateless FastAPI service that can run as multiple instances; database on managed Supabase.	Scale up by adding Render instances and a larger Supabase plan
3	Security & Data Privacy	Secrets in environment variables, HTTPS on Vercel and Render, request validation with Pydantic, .env excluded by .gitignore.	No keys in the repository or frontend bundle
4	Reliability & Availability	Managed hosting; backend handles LLM or database errors with clear messages. Free tier sleeps when idle.	Target 99% availability on paid tier; first request after idle up to ~60 s on free tier
5	Usability & Accessibility	Single-page responsive layout, labelled inputs, visible keyboard focus, readable contrast.	Works on mobile and desktop; follows basic WCAG practices (no formal audit)
6	Other	Maintainability: separate frontend and backend, small modules. Portability: runs locally or on any cloud.	New developer can run it from the README in under 15 minutes